

# Joint Probability Distributions

The function  $f(x, y)$  is a **joint probability distribution** or **probability mass function** of the discrete random variables  $X$  and  $Y$  if

1.  $f(x, y) \geq 0$  for all  $(x, y)$ ,
2.  $\sum_x \sum_y f(x, y) = 1$ ,
3.  $P(X = x, Y = y) = f(x, y)$ .

For any region  $A$  in the  $xy$  plane,  $P[(X, Y) \in A] = \sum_A f(x, y)$ .

The function  $f(x, y)$  is a **joint density function** of the continuous random variables  $X$  and  $Y$  if

1.  $f(x, y) \geq 0$ , for all  $(x, y)$ ,
2.  $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = 1$ ,
3.  $P[(X, Y) \in A] = \int \int_A f(x, y) \, dx \, dy$ , for any region  $A$  in the  $xy$  plane.

The **marginal distributions** of  $X$  alone and of  $Y$  alone are

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

for the discrete case, and

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y) \, dx$$

for the continuous case.

**Example 3.14:** Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If  $X$  is the number of blue pens selected and  $Y$  is the number of red pens selected, find

- (a) the joint probability function  $f(x, y)$ ,

Table 3.1: Joint Probability Distribution for Example 3.14

| $f(x, y)$     |   | $x$            |                 |                | Row<br>Totals   |
|---------------|---|----------------|-----------------|----------------|-----------------|
|               |   | 0              | 1               | 2              |                 |
| $y$           | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|               | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|               | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| Column Totals |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |

**Example 3.16:** Show that the column and row totals of Table 3.1 give the marginal distribution of  $X$  alone and of  $Y$  alone.

**Solution:** For the random variable  $X$ , we see that

$$g(0) = f(0, 0) + f(0, 1) + f(0, 2) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14},$$

$$g(1) = f(1, 0) + f(1, 1) + f(1, 2) = \frac{9}{28} + \frac{3}{14} + 0 = \frac{15}{28},$$

and

$$g(2) = f(2, 0) + f(2, 1) + f(2, 2) = \frac{3}{28} + 0 + 0 = \frac{3}{28},$$

which are just the column totals of Table 3.1. In a similar manner we could show that the values of  $h(y)$  are given by the row totals. In tabular form, these marginal distributions may be written as follows:

|        |                |                 |                |
|--------|----------------|-----------------|----------------|
| $x$    | 0              | 1               | 2              |
| $g(x)$ | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ |

|        |                 |               |                |
|--------|-----------------|---------------|----------------|
| $y$    | 0               | 1             | 2              |
| $h(y)$ | $\frac{15}{28}$ | $\frac{3}{7}$ | $\frac{1}{28}$ |

**Example 3.15:** A privately owned business operates both a drive-in facility and a walk-in facility. On a randomly selected day, let  $X$  and  $Y$ , respectively, be the proportions of the time that the drive-in and the walk-in facilities are in use, and suppose that the joint density function of these random variables is

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y), & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Verify condition 2 of Definition 3.9.

(b) Find  $P[(X, Y) \in A]$ , where  $A = \{(x, y) \mid 0 < x < \frac{1}{2}, \frac{1}{4} < y < \frac{1}{2}\}$ .

**Solution:** (a) The integration of  $f(x, y)$  over the whole region is

$$\begin{aligned}\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy &= \int_0^1 \int_0^1 \frac{2}{5}(2x + 3y) \, dx \, dy \\ &= \int_0^1 \left( \frac{2x^2}{5} + \frac{6xy}{5} \right) \Big|_{x=0}^{x=1} dy \\ &= \int_0^1 \left( \frac{2}{5} + \frac{6y}{5} \right) dy = \left( \frac{2y}{5} + \frac{3y^2}{5} \right) \Big|_0^1 = \frac{2}{5} + \frac{3}{5} = 1.\end{aligned}$$

(b) To calculate the probability, we use

$$\begin{aligned}P[(X, Y) \in A] &= P\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2}\right) \\ &= \int_{1/4}^{1/2} \int_0^{1/2} \frac{2}{5}(2x + 3y) \, dx \, dy \\ &= \int_{1/4}^{1/2} \left( \frac{2x^2}{5} + \frac{6xy}{5} \right) \Big|_{x=0}^{x=1/2} dy = \int_{1/4}^{1/2} \left( \frac{1}{10} + \frac{3y}{5} \right) dy \\ &= \left( \frac{y}{10} + \frac{3y^2}{10} \right) \Big|_{1/4}^{1/2} \\ &= \frac{1}{10} \left[ \left( \frac{1}{2} + \frac{3}{4} \right) - \left( \frac{1}{4} + \frac{3}{16} \right) \right] = \frac{13}{160}.\end{aligned}$$

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**Example 3.17:** Find  $g(x)$  and  $h(y)$  for the joint density function of Example 3.15.

**Solution:** By definition,

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy = \int_0^1 \frac{2}{5}(2x + 3y) \, dy = \left( \frac{4xy}{5} + \frac{6y^2}{10} \right) \Big|_{y=0}^{y=1} = \frac{4x + 3}{5},$$

for  $0 \leq x \leq 1$ , and  $g(x) = 0$  elsewhere. Similarly,

$$h(y) = \int_{-\infty}^{\infty} f(x, y) \, dx = \int_0^1 \frac{2}{5}(2x + 3y) \, dx = \frac{2(1 + 3y)}{5},$$

for  $0 \leq y \leq 1$ , and  $h(y) = 0$  elsewhere.

Let  $X$  and  $Y$  be two random variables, discrete or continuous, with joint probability distribution  $f(x, y)$  and marginal distributions  $g(x)$  and  $h(y)$ , respectively. The random variables  $X$  and  $Y$  are said to be **statistically independent** if and only if

$$f(x, y) = g(x)h(y)$$

for all  $(x, y)$  within their range.

**Example 3.21:** Show that the random variables of Example 3.14 are not statistically independent.

**Proof:** Let us consider the point  $(0, 1)$ . From Table 3.1 we find the three probabilities  $f(0, 1)$ ,  $g(0)$ , and  $h(1)$  to be

$$\begin{aligned} f(0, 1) &= \frac{3}{14}, \\ g(0) &= \sum_{y=0}^2 f(0, y) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14}, \\ h(1) &= \sum_{x=0}^2 f(x, 1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}. \end{aligned}$$

Clearly,

$$f(0, 1) \neq g(0)h(1),$$

and therefore  $X$  and  $Y$  are not statistically independent. ┐

| Distribution            | Discrete                                                    | Continuous                                                    |
|-------------------------|-------------------------------------------------------------|---------------------------------------------------------------|
| Marginal distributions  | $P(x) = \sum_y P(x, y)$<br>$P(y) = \sum_x P(x, y)$          | $f(x) = \int f(x, y) dy$<br>$f(y) = \int f(x, y) dx$          |
| Independence            | $P(x, y) = P(x)P(y)$                                        | $f(x, y) = f(x)f(y)$                                          |
| Computing probabilities | $P\{(X, Y) \in A\}$<br>$= \sum \sum_{(x, y) \in A} P(x, y)$ | $P\{(X, Y) \in A\}$<br>$= \iint_{(x, y) \in A} f(x, y) dx dy$ |

TABLE 4.2: Joint and marginal distributions in discrete and continuous cases.

## Conditional Distributions:

Let  $X$  and  $Y$  be two random variables, discrete or continuous. The **conditional distribution** of the random variable  $Y$  given that  $X = x$  is

$$f(y|x) = \frac{f(x, y)}{g(x)}, \text{ provided } g(x) > 0.$$

Similarly, the conditional distribution of  $X$  given that  $Y = y$  is

$$f(x|y) = \frac{f(x, y)}{h(y)}, \text{ provided } h(y) > 0.$$

**Example 3.18:** Referring to Example 3.14, find the conditional distribution of  $X$ , given that  $Y = 1$ , and use it to determine  $P(X = 0 | Y = 1)$ .

**Solution:** We need to find  $f(x|y)$ , where  $y = 1$ . First, we find that

$$h(1) = \sum_{x=0}^2 f(x, 1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}.$$

Now

$$f(x|1) = \frac{f(x, 1)}{h(1)} = \left(\frac{7}{3}\right) f(x, 1), \quad x = 0, 1, 2.$$

Therefore,

$$f(0|1) = \left(\frac{7}{3}\right) f(0, 1) = \left(\frac{7}{3}\right) \left(\frac{3}{14}\right) = \frac{1}{2}, \quad f(1|1) = \left(\frac{7}{3}\right) f(1, 1) = \left(\frac{7}{3}\right) \left(\frac{3}{14}\right) = \frac{1}{2},$$

$$f(2|1) = \left(\frac{7}{3}\right) f(2, 1) = \left(\frac{7}{3}\right) (0) = 0,$$

and the conditional distribution of  $X$ , given that  $Y = 1$ , is

|          |               |               |   |
|----------|---------------|---------------|---|
| $x$      | 0             | 1             | 2 |
| $f(x 1)$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 0 |

Finally,

$$P(X = 0 | Y = 1) = f(0|1) = \frac{1}{2}.$$

Therefore, if it is known that 1 of the 2 pen refills selected is red, we have a probability equal to  $1/2$  that the other refill is not blue. 

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**Example 3.19:** The joint density for the random variables  $(X, Y)$ , where  $X$  is the unit temperature change and  $Y$  is the proportion of spectrum shift that a certain atomic particle produces, is

$$f(x, y) = \begin{cases} 10xy^2, & 0 < x < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Find the marginal densities  $g(x)$ ,  $h(y)$ , and the conditional density  $f(y|x)$ .
- (b) Find the probability that the spectrum shifts more than half of the total observations, given that the temperature is increased by 0.25 unit.

**Solution:** (a) By definition,

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) \, dy = \int_x^1 10xy^2 \, dy \\ &= \frac{10}{3} xy^3 \Big|_{y=x}^{y=1} = \frac{10}{3} x(1 - x^3), \quad 0 < x < 1, \\ h(y) &= \int_{-\infty}^{\infty} f(x, y) \, dx = \int_0^y 10xy^2 \, dx = 5x^2 y^2 \Big|_{x=0}^{x=y} = 5y^4, \quad 0 < y < 1. \end{aligned}$$

Now

$$f(y|x) = \frac{f(x, y)}{g(x)} = \frac{10xy^2}{\frac{10}{3}x(1 - x^3)} = \frac{3y^2}{1 - x^3}, \quad 0 < x < y < 1.$$

(b) Therefore,

$$P\left(Y > \frac{1}{2} \mid X = 0.25\right) = \int_{1/2}^1 f(y \mid x = 0.25) \, dy = \int_{1/2}^1 \frac{3y^2}{1 - 0.25^3} \, dy = \frac{8}{9}. \quad \blacksquare$$

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**Example 3.20:** Given the joint density function

$$f(x, y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, \ 0 < y < 1, \\ 0, & \text{elsewhere,} \end{cases}$$

find  $g(x)$ ,  $h(y)$ ,  $f(x|y)$ , and evaluate  $P(\frac{1}{4} < X < \frac{1}{2} \mid Y = \frac{1}{3})$ .

**Solution:** By definition of the marginal density. for  $0 < x < 2$ ,

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) \, dy = \int_0^1 \frac{x(1+3y^2)}{4} dy \\ &= \left( \frac{xy}{4} + \frac{xy^3}{4} \right) \Big|_{y=0}^{y=1} = \frac{x}{2}, \end{aligned}$$

and for  $0 < y < 1$ ,

$$\begin{aligned} h(y) &= \int_{-\infty}^{\infty} f(x, y) \, dx = \int_0^2 \frac{x(1+3y^2)}{4} dx \\ &= \left( \frac{x^2}{8} + \frac{3x^2y^2}{8} \right) \Big|_{x=0}^{x=2} = \frac{1+3y^2}{2}. \end{aligned}$$

Therefore, using the conditional density definition, for  $0 < x < 2$ ,

$$f(x|y) = \frac{f(x, y)}{h(y)} = \frac{x(1+3y^2)/4}{(1+3y^2)/2} = \frac{x}{2},$$

and

$$P\left(\frac{1}{4} < X < \frac{1}{2} \mid Y = \frac{1}{3}\right) = \int_{1/4}^{1/2} \frac{x}{2} dx = \frac{3}{64}.$$

